AeroTrust at a Glance

Avionics Telemetry Integrity Observatory & Verification System

`Tag: v0.3.2-verified`

`Hash: b6011043368eca73`

SYNTHETIC DATASET NOTICE: All trajectories and threat injections are synthetic simulations. Numbers derived from pooled clean seeds (3001-3003; 18,000 scheduled legs). No operational commercial flights are depicted.

1. EXECUTIVE SUMMARY

AeroTrust is an avionics data-integrity observatory designed to evaluate aircraft ADS-B telemetry against GNSS spoofing, interference, and atmospheric radiation anomalies in a single unified interface. Operating under frozen verification benchmarks on synthetic flight trajectories, it pairs physics-based altimetry consistency checks with an unsupervised abstention engine, classifying telemetry into four standardized evidence tiers (`no_alert` , `flagged_altitude_inconsistency` , `jamming_degraded` , `insufficient`). It features an aggregate Silent Acceptance Rate (SAR) auditor, geographic H3 interference mapping, and an Incident Clock scoping calculator for aviation airworthiness directives.

2. OPERATIONAL CAPABILITIES & BOUNDARIES

✓ WHAT AEROTRUST CAN DO

- **Altimetry Verification:** Compares barometric vs GNSS altitude residuals, flagging vertical discrepancies > 120.0 m (Check a).
- **Interference Clustering:** Aggregates multi-aircraft NIC/NACp/SIL drops into H3 spatial cells for regional jamming detection.
- **Evidence-Tiered Output:** Never exposes uncalibrated probabilities or claims flights are "safe"; uses standardized evidence tiers.
- **Privacy-Preserving SAR:** Evaluates Silent Acceptance Rate with cell suppression for cohorts where N < 30.
- **Incident Clock:** Models fleet scoping delay sensitivity under airworthiness directives using transparent parameters.

✗ WHAT AEROTRUST CANNOT DO

- **0% Horizontal Spoofing Recall:** Cannot detect lateral track shifts or circular waypoint attacks (vertical only).
- **No Real-World Certification:** Benchmarks are synthetic; not calibrated on certified Flight Data Recorders.
- **Consensus Inactive:** Multi-aircraft consensus (Check d) is disabled due to self-referential graph loops.
- **Peacetime Base-Rate Drop:** At 0.01% threat prevalence, precision drops to 0.2% due to background barometric noise.
- **No Radiation Causality:** Solar proton flux and Kp indices serve strictly as illustrative exposure context, not a cause.
- **No Audited Financials:** Incident Clock cases require verified primary AD documents (currently unchecked).

3. VERIFIED BENCHMARK EVIDENCE & PERFORMANCE METRICS

Metric / Capability	Verified Result	Baseline & Operational Evaluation Scope	Tag & Status
Spoofing Precision (Check a)	**83.0%** (79.7% – 85.8%)	Test Base Rate = 17.47% (5,225 segmented tracks)	`v0.3.2-verified` (Frozen)
Spoofing Recall (Check a)	**40.8%** (37.6% – 44.0%)	Altimetry consistency detector (|Geo - Baro| > 120 m)	`v0.3.2-verified` (Frozen)
Horizontal Spoofing Recall	**0.0%**	Lateral displacements produce 0 altitude residual	`v0.3.2-verified` (Known Limit)
Clean False Positives (Legs)	**42 to 52** per seed	~35-36 baro-bias draws + 7-16 boundary leg splits	`v0.3.2-verified` (Documented)
Multi-Aircraft Consensus	**0.0% (Inactive)**	Check (d) disabled; needs re-validation in v0.4	`v0.3.1-claims` (Disabled)
SAR Suppression Threshold	**N ≥ 30 enforced**	Sub-threshold cells suppressed; no marginal leakage	`v0.3.2-verified` (Enforced)

4. PLAIN-LANGUAGE DISCLAIMERS & KNOWN WEAKNESSES

1. **Altitude-consistency only:** Only vertical discrepancies trigger detection.
2. **False alarm base rate:** In peacetime airspaces, sensor turbulence produces ~117 false alarms per 10,000 clean flights.
3. **Abstention scope:** Benchmark commercial flights have N ≥ 68 points; abstention is exercised only in the demo dataset.
4. **Incident case presets:** All 3 presets (A320 Solar, A320 Panels, A330neo) display "primary source missing" in red until primary regulatory airworthiness directives are formally verified.